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Atom Optics and Quantum Technologies
June 16th, 2025

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Part 1

A beam of ^{87}Rb atoms is emitted from an oven with initial velocity v_0 along the z-axis. To decelerate and stop the atoms, a counter-propagating laser beam is applied along the same axis. Assume that the atoms experience a constant deceleration a_0 due to the radiation pressure force. The atom enters the slowing region at position $z = 0$ with velocity v_0, and is brought to rest at $z = L$.

a) Draw the energy level diagram of ^{87}Rb including fine structure, hyperfine structure, and Zeeman splitting up to the $5^2P_{3/2}$ state. Pay attention to the correct ordering of the Zeeman sublevels.
b) Indicate with an arrow the cycling transition of ^{87}Rb, knowing that the initial state is $5^2S_{1/2}$ $(F' = 2, M_{F'} = +2)$ and the laser has right-hand circular polarization. Denote the final state as $(F'', M_{F''})$.
c) Derive an expression for the Doppler shift $\Delta\omega_D(z)$ (variation of resonance frequency of the atom), for $0 < z < L$. Compute the maximum value of the Doppler shift in units of Hz.
d) Derive the expression for the magnetic field $B(z)$ required to maintain resonance between the atom and the laser at all positions z, taking into account both the ground and excited state Zeeman shifts. Compute the maximum value of the magnetic field in units of T.
e) Assume the atom starts at an initial velocity v_0 = 300 m/s. Calculate the minimum length L of the slowing region required to bring the atom to rest.
f) Calculate the total number of photons that must be scattered by the atom in order to stop it.
g) Discuss qualitatively what happens to atoms whose initial velocities are slightly smaller or slightly larger than v_0. Will they still be efficiently slowed down?

Constants and data

Nuclear spin: $I = 3/2$
$g_{F'}(5^2S_{1/2}, F' = 2) \approx 0.5$
$g_{F''}(5^2P_{3/2}, F'' = 3) \approx 0.67$
$\lambda = 780.241$ nm
$\Gamma = 2\pi \times 6.07$ MHz
$\mu_B = 9.274 \times 10^{-24}$ J/T
$h = 6.626 \times 10^{-34}$ J·s
Atomic mass of ^{87}Rb: $m = 1.443 \times 10^{-25}$ kg

Part 2

Magneto-Optical-Trap (MOT)

a) Explain qualitatively the working principle a MOT, and provide arguments supporting the existence of a stationary (or equilibrium) velocity;

Part 3

The quantum Langevin equation of motion for an empty cavity mode is:

$$\dot{\hat{a}} = -(k + i\Delta)\hat{a} + \sqrt{2k_1}\hat{A}_{in,1} + \sqrt{2k_2}\hat{A}_{in,2} + \sqrt{2k_L}\hat{A}_{in,L}$$

where $\hat{a}$ is the operator associated to the intracavity mode, and $\hat{A}_{in,1}, \hat{A}_{in,2}, \hat{A}_{in,L}$ are the continuous variable operator associated to modes entering the cavity at mirrors and losses (for simplicity, an idealized mirror model has been used), k_1, k_2, k_L are logarithmic losses normalized to the cavity round-trip time, $k = k_1 + k_2 + k_L$ is the overall decay rate of the cavity mode, and Δ is the frequency detuning between the cavity mode and cavity resonance of order n. Consider now, the special case where the frequency detuning and cavity losses are negligible and cavity mirrors are matched ($\Delta = 0, k_1 = k_2, k_L \cong 0$).

a) Derive the expressions for the variances of the two quadratures $V1_{out,1}(f), V2_{out,1}(f)$ of the mode at the output of mirror 1.
b) Derive the limits of the two variances $V1_{out,1}(f), V1_{out,2}(f)$ at low and high Fourier frequencies.
c) Draw a graph of the variances of the quadrature amplitudes of the output modes from mirrors 1 and 2, $V1_{out,1}(f), V1_{out,2}(f)$ assuming that the cavity is seeded on mirror 1 by a laser with $V1_{in,1}(f)$=2.